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MSBA Capstone Project

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Executive Summary

Type your executive summary here.

Remember, the executive summary can vary widely from project to project. Keep it concise and focus on the information that your reader will care about most. Use subheadings, bullet points, and graphics wherever applicable.

Remember that many successful executive summaries begin with a problem/solution section, in which the client's problem and the business analytics solution are succinctly described.

Company Overview

Type your company overview here.

This section should be a concise, inspiring, and forward-looking description of your client organization and the business unit you are focusing on. Explain the products or services, their business model, and competitive advantages.

Team and Project

Project Description

Describe the analytics project in terms of the opportunity or challenge, the specific functional areas, topics, vocabulary, analysis methods, and skills needed.

Contextualize the analytics opportunity and overall business scenario first, and then drill down to the specific functional areas to demonstrate your command for the overall challenge provided.

Team Name and Organization

Name your team, and type brief bios and roles of the key leaders of your analytics consulting team here.

Skills Concerns

List your team's skill gaps here, if any, and explain how you will compensate for them.

Strategic Analysis

External Analysis

This is where you include opportunities and threats emerging from the external environment, including information like the industry growth rate, number and size of competitors, and target markets. Be sure to include a list of your sources in the references.

Internal Analysis

Type the internal strategic analysis for your client here in terms of their strengths, core competencies, areas of improvement, key resources, and sources of competitive advantages. Be sure to include a list of your sources in the references.

Analytics Opportunity

Here is where you will describe how you explored and identified the analytics opportunity and defined the overall data and analytics strategy for your client. In addition to the strategic analysis you conducted in the previous section, some questions below can help get you started thinking in this direction:

- *What are your client's key business drivers?*
- *What are the opportunities data and analytics can unlock for your clients?*
- *What are your client's pain points in their business processes?*
- *What are the business impacts of these pain points?*
- *What are the key causes of the pain points, and how can data provide alternate solutions?*

Rationale

Explain the key problem and the analytics solution your team is proposing. Provide details on how your ideas can be applied to resolve the identified problem and why this solution will work.

Stakeholder and Requirement Analysis

Key Stakeholders

Describe the key stakeholders of the analytics projects for your client here. In addition, also discuss the process flow.

Requirements

Discuss the 5-10 examples of questions, requirements or analytical procedures that your different stakeholders may be interested in. Articulate specific expectations each stakeholder may have from the analytics solution your team is proposing.

Success Criteria

After conducting the stakeholder and requirement analysis, you and your client should have clarity about the main objectives of the analytics project. Here you'll describe further what a successful outcome of your analytics project will look like in more concrete terms. You'll define the key performance indicators relevant to your analytics projects.

You collaborate with your team and your client, seek feedback from the academic team, and sharpen your analysis and ideas. Then you propose and agree upon the success criteria on both the business side and technical side. These criteria should be measurable and specific.

Scope and Schedule of Deliverables

Feasibility

In this phase, you'll assess the data and resource needs for your analytics project, and then conduct the technical, economic, and operational feasibility analysis for your project, and share your findings.

Scope

Feasibility analysis will provide further insights into understanding your proposed analytics project's size, structure, and complexity. This will help you get a much clearer picture of the problem and define the scope in terms of articulating what your possible solution could deliver in the defined time frame.

Deliverables and Schedule

After you define the scope, you gain agreement on the key deliverables and the project schedule and describe that here. Your team will also lay out your plan for interim deliverables through your capstone journey, which will be used as a basis for your regular updates. These will also be used to set your client's and stakeholders' expectations. If you are working with a real client, they should sign off on your proposed success criteria, schedule, and deliverables.

Data Understanding

Describe different variables present in the data, and share your overall observation about the available datasets.

Data Preparation

Explain how you need to prepare the data for modeling and analysis. Describe how you cleaned the data, and discuss the data quality. Discuss the missing values and connecting fields if multiple datasets are used.

Here, you'll also discuss if you need to create a separate custom database as a part of your overall analytics solution. If so, what will be the schema and the data models, and discuss your plans of how you will extract, transform and load the data. Eventually, this will affect how analysts run the code for the analytical procedures and models you designed and how executives review the results.

Exploratory Data Analysis

Conduct a thorough exploratory data analysis to generate deep insights into data and the overall context of the business, and share your findings here. Also, discuss any data treatments and transformations you conducted in this phase, along with their rationale. Determine if the given data is sufficient, and share a list of follow-up questions for your client. Also, share the Python/R code link, and Tableau visualizations link to your exploratory data analysis. We recommend you conduct this analysis using Python or R. Depending on your team's proficiency with Python/R; please feel free to use Tableau as you need to substantiate the insights.

Methods and Frameworks

Here you'll discuss the method you chose for conducting the research and finding answers to the key questions posed or the alternate solutions to the key challenges faced by the clients.

Overall Approach

You can pick a combination of descriptive, predictive, or prescriptive analytics methods and techniques. This is your opportunity to apply what you are learning in the concentrations, communicating with data, analytics methods and frameworks, analytics with big data and AI, blockchain, or advanced analytics methods and frameworks, depending on the analytics challenge your client is facing, their requirements and your team's preferences. Describe your clear overall step-by-step approach.

Tools and Technologies

Provide all the details and rationales for choosing specific analytics methods, tools, and techniques. provide step-by-step approaches.

Alignment with Project Goals and Deliverables

At this stage, it is also a good idea to take a step back to research questions, goals, and business challenge pain points if they are resolved with your method and approach. Also important to step back and assess and align your methodology with your identified and agreed-upon deliverables and schedules.

Architect the Solution: Proof of Concept

Describe the analytics workflow for the proposed analytics solution, with a thorough explanation of your work, process steps, formulas, calculations, code generated, and output and the results. Also, explain how you captured the analytics opportunity you identified through your proposed solution, in the short term and also in the long run with a review and resolution of the success criteria identified in the earlier section.

Your client may also like to develop a proof of concept (POC) before fully committing to the idea. POC is generally seen as a miniature project to demonstrate the value and feasibility of the overall analytics solution designed. If your client is trying something new, this way, you can help them explore a way forward, discover pitfalls or provide confirmation that they are on the right track.

Evaluate Business Value: Cost Analysis

Here you'll explain the impact and business value of the analytics solution your team has developed. For this, you need to estimate both the overall costs and benefits of the proposed solution.

Capital Requirements

List capital requirements for your business analytics project, as well as any funds that have already been raised. Explain how these funds will be used.

Summary of Cost Analysis

Consider including a table that summarizes the projected costs and benefits of your proposed solution. [Here](#) is some more guidance on how to do so.

Assumptions

List the assumptions that underpin your financial projections. Explain how you arrived at your assumptions and show that they are in line with industry expectations.

ROI Analysis

Explain how long you have projected it will take for your business to get a return on your initial investments.

Risk & Reward Analysis

Describe the expected rate of return and the likelihood of achieving that rate.

Scale up the POC: Recommendations

In this section, you, as a team of analytics consultants, will draft alternate options to capture the identified analytics opportunity and draft your recommendations.

If your client opts for developing POC, you provide further guidance on how to scale and what challenges to look out for in this process. It's your job to explain what needs to happen for your solution to be “scalable” and specify what “scalable” means.

References

List all your sources here. Use [Chicago-style formatting](#).

Appendix

Add relevant code links, financial projections, datasets links, additional deliverables, optional material, etc. here